

2.3 Gauss's Law: Cylindrical and Planar Symmetry

Date: _____

1. An infinitely long line of charge has linear density $\lambda = 2.0 \mu\text{C}/\text{m}$. Find E at a perpendicular distance of 0.050 m.

2. A large flat insulating sheet carries surface charge density $\sigma = 1.0 \times 10^{-6} \text{ C/m}^2$. Find the field near the sheet.
3. Explain why the field just outside a charged conductor in electrostatic equilibrium is $E = \sigma/\epsilon_0$, twice the value for the thin sheet above.